Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find |0| and describe it as a distance.

2. Find the opposite of 9.

3. Reflect (-3, 4) across the y-axis.

4. Find |−12| and describe it as a distance.

5. Solve 6x = 54.

6. Solve 1.5x = 4.80.

7. A square pyramid has base side 4 cm and triangular-face height 3 cm. Find surface area.

8. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

9. Find 25% of 24 stickers.

10. Ben says −10 > −3 because 10 > 3. Use a number line to explain and repair his statement.

11. Write signed numbers for 9 meters below sea level and 4 meters above it. State what zero represents.

12. Order these values from least to greatest: $\frac{3}{2}$, -9, 0, -2.

13. Find the opposite of −9, then its opposite. Explain why you return to the start.

14. Find |−9|, |9|, and |0|. Compare the first two.

15. Which is warmer, −9°C or −11°C? Which has greater absolute value?

16. Find the distance between −9 and 4 on a number line. Split the distance at zero.

17. Name the quadrant of each point: A(4, 9), B(−4, 9), C(−4, −9), D(4, −9).

18. Reflect (−4, 9) across the x-axis and then, starting again, across the y-axis.

CHALLENGE / Explain your plan and check the result

C1. A number-line game allows integers x with $-4 \le x \le 3$. A bonus requires $|x| \ge 3$. List all bonus numbers and their opposites. Which opposites remain allowed in the original game?

C2. A square has consecutive vertices (−2, −2), (3, −2), (3, 3), (−2, 3). Find its perimeter. Reflect the square across the y-axis. What is the leftmost x-coordinate of the image? Explain why a vertex originally left of the axis can move right of it.